

VGA GRAPHICS DEMO

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Modules and Architecture

vga_top - connects all modules and finalizes the architecture

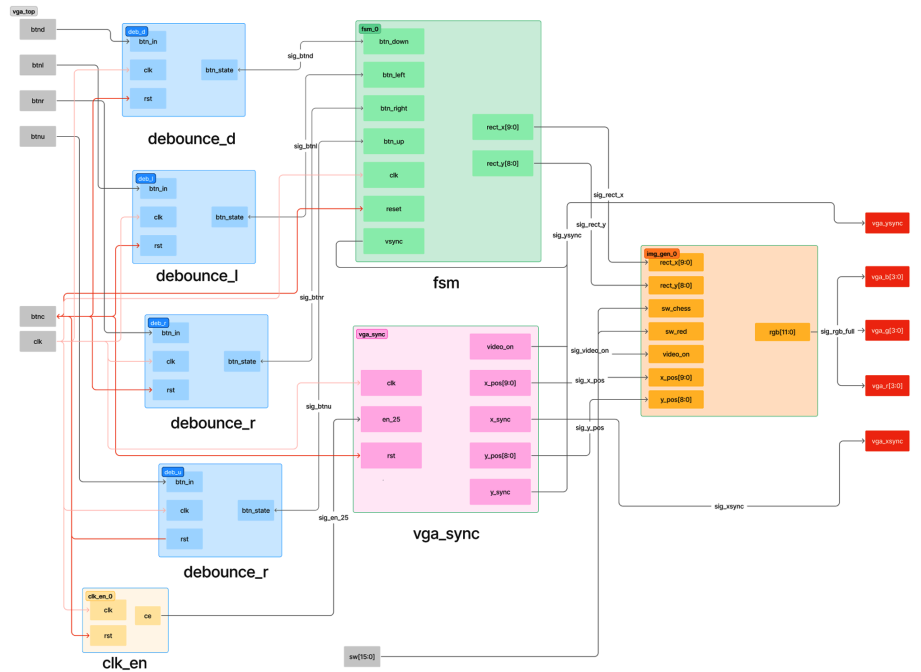
fsm - manages the logic for moving on the screen, based on inputs and vertical sync

img_gen - renders a rectangle, colors switching and background

vga_sync - generates horizontal and vertical timing signals for 640x480 display

debounce - ensures one physical press equals exactly one clean digital signal

clk_en - transforms system clock into a periodic pulse, tells other modules when to perform an action



I/O and controls

btnd - moves the square down along the Y-axis

btlu - moves the square up along the Y-axis

btlr - moves the square left along the X-axis

btrr - moves the square right along the X-axis

btnc - instantly resets the square's position to the center of the screen

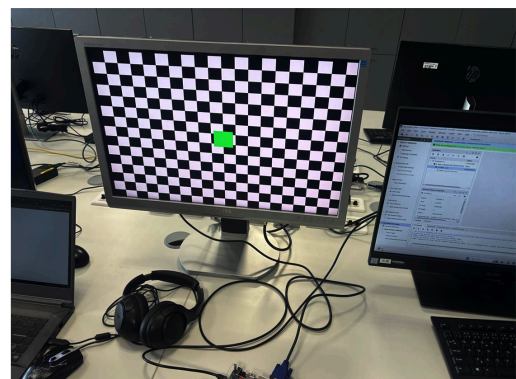
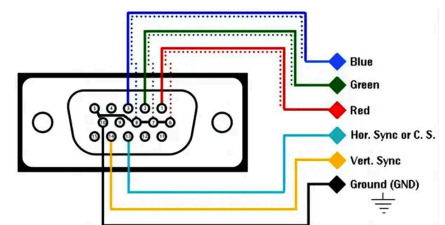
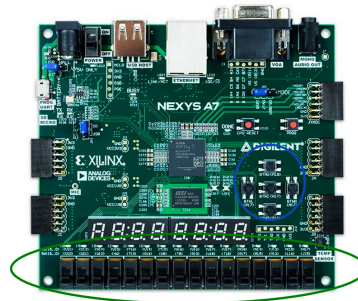
sw[15:0] - provides physical switches on the board, turning sw[0] changes the color of the rectangle and sw[15] changes the background

clk - the main system clock that drives all synchronous logic in the design

vga_xsync - tells the monitor to go back to the left edge of the screen, it's also used to move to the next row of pixels once per line

vga_ysync - tells the monitor to go back to the top of the screen, it's also used to update the animation once per frame

vga_r/g/b - physical pins sending analog color data to the VGA port



What does it display?

It displays a rectangle which can be moved based on the controls.

We can switch the colors of the rectangle by turning on/off switches

Where the last switch changes the background to a chess pattern

<https://github.com/afflorld/DE1>